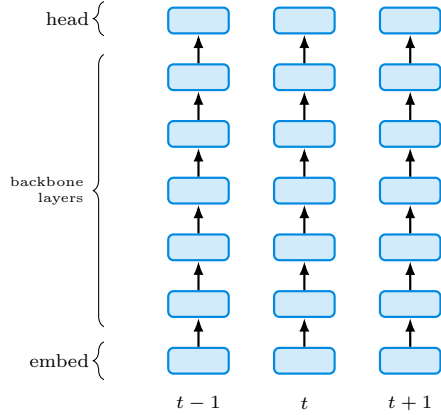


Training-time information flows

Depth recurrence OFF

Vanilla transformer

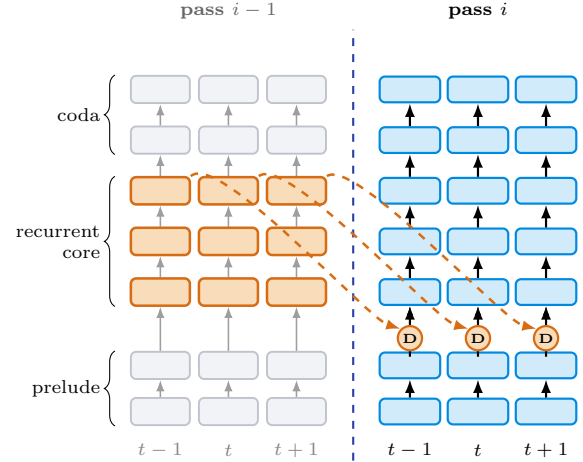
A uniform layer stack: one embedding layer, backbone layers, and one head layer.



Depth recurrence ON

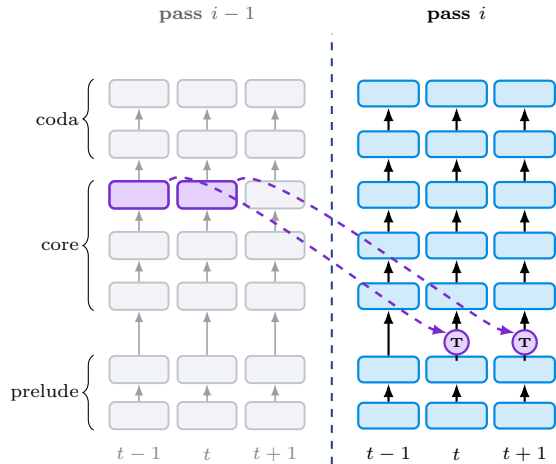
Depth-recurrent / looped LM

Recurrent-core layers emit depth state in pass $i-1$; pass i reads it before the recurrent-core band.



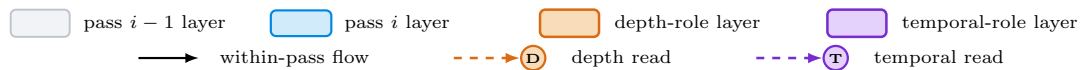
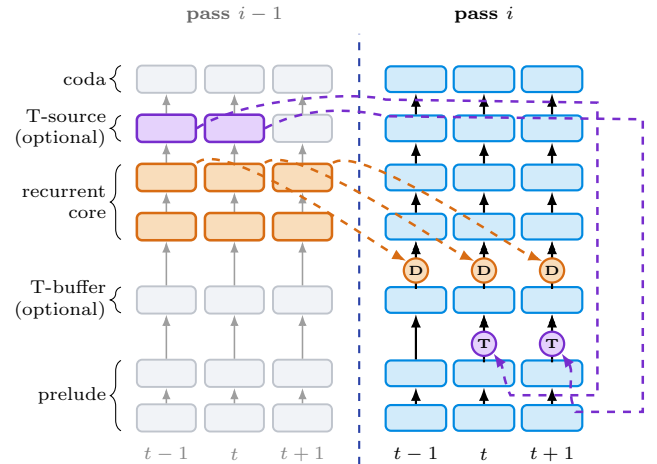
Temporally recurrent

The last core layer emits temporal state in pass $i-1$; pass i reads it before the core band.



Depth + temporal hybrid

Distinct roles: T-buffer, recurrent core, T-source, and coda. Temporal routing comes from T-source; depth routing comes from the top recurrent core layer.



Temporal recurrence OFF

Temporal recurrence ON